

# Phillip W. Yoon

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## EDUCATION

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| <b>Hunter College, City University of New York</b><br><i>M.A. in Mathematics.</i>                                                                                     | 2025–2027 (expected)<br>New York, NY |
| <b>University of Pennsylvania, Graduate School of Education</b><br><i>M.S.Ed. in Independent School Teaching Residency</i>                                            | 2022–2024<br>Philadelphia, PA        |
| <b>Northern Illinois University</b><br><i>Graduate (M.A.) coursework in philosophy, 18 credits (non-degree)</i>                                                       | 2021–2022<br>DeKalb, IL              |
| <b>Johns Hopkins University</b><br><i>B.A. in Mathematics and Philosophy, General Honors.</i><br><i>Study abroad: Budapest Semesters in Mathematics, Summer 2019.</i> | 2017–2021<br>Baltimore, MD           |

## RESEARCH INTERESTS

Commutative algebra, particularly homological and combinatorial aspects of free resolutions and related invariants; algebraic combinatorics; and noncommutative algebra. Additional interests include topological methods in algebra, computational algebra, and representation-theoretic combinatorics.

## PUBLICATIONS

1. Noah Ripke and **Phillip Yoon**, “Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables,” *Journal of Pure and Applied Algebra* **230** (2026), 108354. [arXiv:2607.10639](https://arxiv.org/abs/2607.10639).

## RESEARCH EXPERIENCE

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| <b>NSF REU/RET in Mathematics, California State University, Chico</b><br><i>Researcher, Commutative Algebra and Combinatorics Group</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | June–July 2026<br>Chico, CA          |
| <ul style="list-style-type: none"><li>• Under the supervision of Guillermo Alesandróni, proved with a coauthor that the multigraded, graded, and total Betti numbers of monomial ideals in five variables are independent of the characteristic of the base field; led the drafting and revision of the resulting manuscript.</li><li>• Developed substantial portions of the homological and combinatorial argument using squarefree reductions, Hochster’s formula, and simplicial homology; wrote Python code reducing 7,580 squarefree ideals to 210 representatives up to variable relabeling and carried out computational verification in Macaulay2.</li><li>• Continuing this work by studying characteristic-specific thresholds for Betti-number dependence through torsion in simplicial homology, including explicit eight-vertex constructions exhibiting odd-primary torsion.</li></ul> |                                      |
| <b>Hunter College, City University of New York</b><br><i>Independent Research in Combinatorics on Words and Noncommutative Algebra</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | October 2025–present<br>New York, NY |
| <ul style="list-style-type: none"><li>• Conducting joint research with Be’eri Greenfeld on growth and nilpotent ideals in associative and monomial algebras through combinatorics on infinite words.</li><li>• Constructed a two-generated connected graded monomial algebra of finite Gelfand–Kirillov dimension with no inclusion-maximal nilpotent two-sided ideal, using combinatorial restrictions on subwords to control growth and ideal structure; continuing to investigate sharper growth constructions.</li></ul>                                                                                                                                                                                                                                                                                                                                                                          |                                      |
| <b>Polymath Jr., University of Minnesota</b><br><i>Research Participant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | June–August 2025<br>Remote           |
| <ul style="list-style-type: none"><li>• Investigated partition functions and recurrence operators for a metaplectic lattice model related to quantum superalgebras under the mentorship of Benjamin Brubaker.</li><li>• Contributed explicit partition-function computations and analysis of local recurrence relations and their representation-theoretic structure; presented this work at the 2026 Joint Mathematics Meetings.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                      |

## RESEARCH TALKS AND PRESENTATIONS

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| <i>“Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables”</i>                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026–2027 |
| <ul style="list-style-type: none"><li>• <b>Invited talk</b>, AMS Special Session on Advances in Commutative Algebra and Connections with Geometry and Combinatorics, Joint Mathematics Meetings, Chicago, IL, January 2027.</li><li>• <b>Invited seminar talk</b>, CUNY Commutative Algebra and Algebraic Geometry Seminar, The Graduate Center, CUNY, New York, NY, November 2026.</li><li>• <b>Contributed talk</b>, AMS Fall Eastern Sectional Meeting, George Washington University, Washington, DC, October 2026.</li></ul> |           |

- **Invited talk**, Route 81 Conference on Commutative Algebra and Algebraic Geometry, Syracuse University, Syracuse, NY, September 2026.  
*Joint work with Noah Ripke.*
- “What Does Small Growth Mean for Algebras?” April 2026  
Hunter Mathematics and Statistics Colloquium: Student Research Presentations, Hunter College, New York, NY.
- “Lattice Models for Quantum Superalgebras: Color and Supercolor” January 2026  
Joint Mathematics Meetings, Washington, DC.
- “Finding Equations for Parametrized Curves of an Affine Variety” May 2025  
**Poster presentation**, CUNY Directed Reading Program, The Graduate Center, CUNY, New York, NY.

## ADDITIONAL MATHEMATICAL EXPERIENCE

- CUNY Directed Reading Program, The Graduate Center** January–May 2025  
*Algebraic Geometry and Computational Commutative Algebra* New York, NY
- Read selected material from Cox, Little, and O’Shea’s *Ideals, Varieties, and Algorithms* with graduate mentor Valeriy Sergeev; completed an expository project and final presentation.

## TEACHING EXPERIENCE

- Hunter College, City University of New York** August 2026–present  
*Adjunct Lecturer* New York, NY  
*Instructor of record: Precalculus (MATH 12550).*
- Columbia Grammar and Preparatory School** August 2025–August 2026  
*Teacher of Mathematics* New York, NY  
*Courses: AP Calculus AB, Precalculus.*
- New York City College of Technology, CUNY** August–December 2024  
*Adjunct Lecturer* Brooklyn, NY  
*Courses: College Algebra and Trigonometry, Precalculus.*
- Trinity School** August 2022–June 2024  
*Mathematics Teaching Fellow* New York, NY  
*Courses: Prealgebra, Honors Algebra I.*
- Johns Hopkins University** August 2018–May 2021  
*Teaching Assistant and Tutor* Baltimore, MD  
*Courses: Discrete Mathematics, Probability and Statistics, Calculus II, Linear Algebra.*

## AWARDS AND HONORS

- John Arthur Loustau Scholarship, Hunter College, CUNY, 2026.
- Gail J. McGovern Scholarship, Johns Hopkins University, 2018–2020.
- Honorable Mention, International Olympiad on Astronomy and Astrophysics, 2016.

## SERVICE AND OUTREACH

- Member, AI Committee, Columbia Grammar and Preparatory School, 2025–2026.
- MATHCOUNTS Coach, Trinity School, 2022–2024.
- Co-organizer, Johns Hopkins Mathematics Tournament, 2018–2020.

## LANGUAGES AND TECHNICAL SKILLS

**Languages:** English, Korean, Japanese.  
**Technical:**  $\LaTeX$ , Python, Macaulay2.

## ADDITIONAL PROFESSIONAL EXPERIENCE

- xAI** January–August 2025  
*AI Tutor* Remote
- Evaluated and improved AI-generated mathematical reasoning and written solutions.